Mohammed Umair Jibran

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Experience

Technology Manager May 2025 – Present
U+ Education | Eight-month Co-op followed by Contract Roles *Richmond Hill, ON*

- Directly managed an **11-member cross-functional team** and supervised **200+ volunteers**, assigning work, reviewing deliverables, and supporting more than **3,000 community service hours**; oversaw **5+** high-school co-op students at a time and delivered **3+** internal development workshops.
- Developed and reviewed features for a Wix/DreamHost website, **volunteer hub application**, and React/Next.js **inventory application** using HTML/CSS, JavaScript, Node.js, PostgreSQL, Vercel, and Postman; translated stakeholder requests into Jira tickets and QA-tested GitHub releases.
- Managed program registrations and operational data in **Excel and monday.com**, cleaning spreadsheets, applying formulas and tools, and building automations for accurate status tracking and reporting.
- Led a **GTA-wide youth-programming analysis** using tens of thousands of Excel records, guiding data analyst volunteers through **Power BI dashboard development** to surface actionable insights for program improvements.

Software Developer Intern Feb 2025 – May 2025
Islamic Books & Souvenirs | Riipen Internship *Toronto, ON*

- Built a Node.js packing-rule engine that joined Shopify-style order JSON with product-dimension CSV data, validated inputs, and generated deterministic packing instructions for **20+ orders**; adopted by the business and contributed to an approximately **15%** combined improvement in order-to-shipment time and packaging costs.

Technology Lead | Executive Director Sep 2024 – Apr 2026
Muslim Student Association at York University *Toronto, ON*

- Developed the front end and server-side Stripe integration for a WisePOS/BBPOS donation workflow, validated terminal data flows, and trained volunteers, supporting more than **$10,000** in fundraising at live events.

Projects

CAN-FD Real-Time ECU Test Bench | *STM32, C/C++, FreeRTOS, CAN-FD, Python*

- Built and operated a two-STM32 hardware bench in C/C++ and FreeRTOS: a sensor ECU transmitted temperature and RPM data while an actuator ECU checked message identity, integrity, and freshness before controlling a fan.
- Implemented queues, interrupts, timers, and buffers for real-time messaging, then developed a Python test harness to capture traffic, inject dropped or delayed frames, and verify timeout handling, fail-safe transitions, and recovery.

RF Asset Finder | *Arduino, C++17, 433 MHz RF, RadioHead, GitHub Actions*

- Developed a tag/base-station protocol and Arduino firmware using device IDs, sequence numbers, CRC-16/CCITT validation, acknowledgements, three-attempt retries, and missing-tag timeouts; added desktop tests for corrupt frames, invalid packet types, duplicate sequences, and timeout state.

SentriCode Security Workbench | *Python, FastAPI, Next.js, TypeScript, Docker, SQLAlchemy*

- Built a local-first security platform that ingests directories, ZIP files, and public GitHub repositories; normalizes source, secret, configuration, and dependency findings and exports JSON, CSV, SARIF, and CycloneDX reports.
- Hardened untrusted-input handling with archive limits, safe extraction, secret redaction, policy gates, scan-history comparisons, and GitHub Actions; verified the release with **142 automated tests**.

Education

York University Toronto, ON
Bachelor of Engineering in Computer Engineering *Expected 2028*

- **Gordon and Agnes (Twambley) Brash Award in Engineering:** Sole engineering student recipient.
- **Relevant Coursework:** Embedded Systems, Communication Networks, Digital Logic Design, Electrical and Electronic Circuits, Computer Organization, Operating Systems, Database Systems, and Engineering Project Management.

Engagement

Selected Speaker: OACAO Aging Well Conference; chosen to represent U+ Education at the provincial conference
Clubs: York University Robotics Club; York University Engineering Society
Professional Associations: IEEE Power & Energy Society (PES); Ontario Society of Professional Engineers (OSPE)

Skills

Programming: C, C++, Python, Java, JavaScript, TypeScript, SQL
Embedded & Hardware: STM32, Arduino, FreeRTOS, CAN-FD, 433 MHz RF, PCB design (Autodesk)
Software & Data: Node.js, React, Next.js, FastAPI, PostgreSQL, MySQL, REST APIs
Testing & Systems: System integration, fault injection, test automation, hardware–software debugging, QA
Tools & Delivery: Git/GitHub, Jira, Postman, Docker, MS Office, monday.com